

JKBOSE 2026

CLASS 11TH

Relations and Functions

MATHS

LECTURE-01

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old NCERT ✗
NEW NCERT ✓
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TOPICS TO BE COVERED

1

Cartesian Product of Two Sets

2

Exercise Discussion 2.1

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Related Problems

Cartesian Products of Sets

let A and B be any two sets, then their Cartesian product is denoted and defined as;

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

For example: * $A = \{1, 2, 3\}$ and $B = \{4, 5\}$

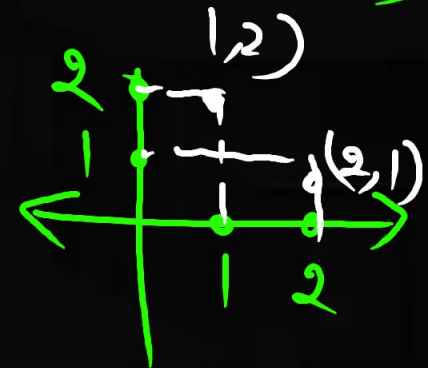
$$\Rightarrow A \times B = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)\}$$

$$\text{Also } B \times A = \{(4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3)\}$$

$$* A = \{a, b\} \text{ \& } B = \{c, d\}$$

$$\Rightarrow A \times B = \{(a, c), (a, d), (b, c), (b, d)\}$$

$$\text{\& } B \times A = \{(c, a), (c, b), (d, a), (d, b)\}$$



Observation:

- $A \times B \neq B \times A$
- $n(A \times B) = n(B \times A) = n(A) \times n(B)$
- No. of subsets of $A \times B = 2^{n(A \times B)} = 2^{n(A) \times n(B)}$

Question

If $A = \{2, 3\}$ and $B = \{1, 2\}$, then $A \times B$ is equal to:

1

$\{(2, 1), (2, 2), (3, 1), (3, 2)\}$.

2

$\{(2, 1), (2, 2), (3, 1), (2, 3)\}$

3

$\{(2, 1), (2, 2), (1, 3), (3, 2)\}$

4

$\{(1, 2), (3, 2)\}$

Question

If $A = \{x \in \mathbb{Z} : (x-2)(x-3) = 0\}$ and $B = \{6, 7\}$ then $A \times B$ is equal to

1

$\{(2, 6), (2, 7), (3, 6), (3, 7)\}$.

2

$\{(1, 6), (2, 6), (3, 6), (3, 7)\}$

3

$\{(1, 6), (9, 7), (9, 6), (9, 7)\}$

4

None of the Above

$$A = \{x \in \mathbb{Z} : (x-2)(x-3) = 0\}$$

$$(x-2)(x-3) = 0$$

$$\Rightarrow x = 2, 3.$$

$$\text{So, } A = \{2, 3\} \text{ \& } B = \{6, 7\}$$

Example

If $(x + 1, y - 2) = (3, 1)$ find the values of x and y .

(Sol) Given that $(x+1, y-2) = (3, 1)$

$$\Rightarrow x+1=3 \text{ \& } y-2=1.$$

$$\Rightarrow x=2, y=3.$$

Question

If $(2x - y, x + y) = (3, 4)$ then the value of **x and y** are respectively equals to :

(Sol) Given $(2x - y, x + y) = (3, 4)$

$$\Rightarrow 2x - y = 3 \text{ --- (1)}$$

$$\& x + y = 4 \text{ --- (2)}$$

Adding (1) & (2), we get;

$$2x - y + x + y = 3 + 4$$

$$\Rightarrow 3x = 7$$

$$\Rightarrow x = \frac{7}{3}$$

Also, $x + y = 4$

$$\Rightarrow y = 4 - x$$

$$\Rightarrow y = 4 - \frac{7}{3}$$

$$\Rightarrow y = \frac{12 - 7}{3} = \frac{5}{3}$$

So, $x = \frac{7}{3}, y = \frac{5}{3}$

☒ 1

3 and 1

☒ 2

1 and 3

☒ 3

2 and 2

☒ 4

None of the Above

$$A \times B \times C = ?$$

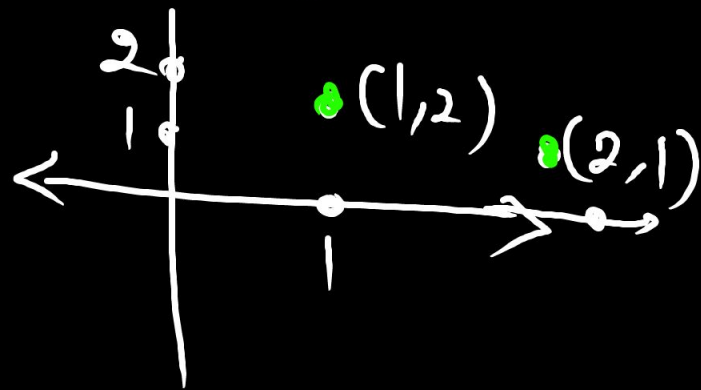
$$(a,b) \neq (b,a)$$



$$* A = \{1,2\}, B = \{3\}, C = \{4,5\}$$

$$\Rightarrow A \times B = \{(1,3), (2,3)\}$$

$$\Rightarrow A \times B \times C = \{(1,3,4), (1,3,5), (2,3,4), (2,3,5)\}$$



Example

Let $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{4, 5, 6\}$. Find

- (i) $A \times (B \cap C)$
- (ii) $(A \times B) \cap (A \times C)$
- (iii) $A \times (B \cup C)$
- (iv) $(A \times B) \cup (A \times C)$

EXERCISE – 2.1

Question

If $\left(\frac{x}{3} + 1, y - \frac{2}{3}\right) = \left(\frac{5}{3}, \frac{1}{3}\right)$, find the values of x and y .

(Sol) Given that $\left(\frac{x}{3} + 1, y - \frac{2}{3}\right) = \left(\frac{5}{3}, \frac{1}{3}\right)$

$$\Rightarrow \frac{x}{3} + 1 = \frac{5}{3}, y - \frac{2}{3} = \frac{1}{3}$$

$$\Rightarrow \frac{x}{3} = \frac{5}{3} - 1, y = \frac{1}{3} + \frac{2}{3}$$

$$\Rightarrow \frac{x}{3} = \frac{5-3}{3}, y = \frac{1+2}{3}$$

$$\Rightarrow \frac{x}{3} = \frac{2}{3}, y = \frac{3}{3}$$

$$\Rightarrow x = 2, y = 1$$

Question

If the set A has 3 elements and the set $B = \{3, 4, 5\}$ then find the number of elements in $(A \times B)$.

$$(Sol) \quad n(A \times B) = n(A) \times n(B)$$

$$\Rightarrow n(A \times B) = 3 \times 3$$

$$\Rightarrow n(A \times B) = 9$$

Question

If $G = \{7, 8\}$ and $H = \{5, 4, 2\}$, find $G \times H$ and $H \times G$.

(Sol) Here $G \times H = \{(7, 5), (7, 4), (7, 2), (8, 5), (8, 4), (8, 2)\}$
 $H \times G = \{(5, 7), (5, 8), (4, 7), (4, 8), (2, 7), (2, 8)\}$.

Question

State whether each of the following statements are true or false. If the statement is false, rewrite the given statement correctly.

(i) If $P = \{m, n\}$ and $Q = \{n, m\}$ then $P \times Q = \{(m, n), (n, m)\}$.

(ii) If A and B are non-empty sets, then $A \times B$ is a non-empty set of ordered pairs (x, y) such that $x \in A$ and $y \in B$.

(iii) If $A = \{1, 2\}$, $B = \{3, 4\}$, then $A \times (B \cap \phi) = \phi$.

$$A \times \phi = \phi.$$

(i) (Sol) False.

$$P \times Q = \{(m, n), (m, m), (n, n), (n, m)\}$$

(ii) (Sol) True.

(iii) True.

Question

$$A = \{-1, 1\}, \quad A = \{-1, 1\}.$$

If $A = \{-1, 1\}$, find $A \times A \times A$.

$$(Sol) \quad A \times A = \{(-1, -1), (-1, 1), (1, -1), (1, 1)\}.$$

$$\Rightarrow A \times A \times A = \{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), \\ (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}.$$

Question

If $A \times B = \{(a, x), (a, y), (b, x), (b, y)\}$. Find A and B.

hp/w.

Question

Let $A = \{1, 2\}$, $B = \{1, 2, 3, 4\}$, $C = \{5, 6\}$ and $D = \{5, 6, 7, 8\}$. Verify that

- (i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
- (ii) $A \times C$ is a subset of $B \times D$.

(Sol) (i) Here $A = \{1, 2\}$

$$\& B \cap C = \phi$$

$$\Rightarrow A \times (B \cap C) = A \times \phi = \phi$$

$$\Rightarrow A \times (B \cap C) = \phi \quad \text{--- (1)}$$

$$\text{Further } A \times B = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4)\}$$

$$\& A \times C = \{(1, 5), (1, 6), (2, 5), (2, 6)\}$$

$$\Rightarrow (A \times B) \cap (A \times C) = \phi \quad \text{--- (2)}$$

From (1) & (2), we get;

$$A \times (B \cap C) = (A \times B) \cap (A \times C).$$

(ii) Here $A \times C = \{(1,5), (1,6), (2,5), (2,6)\}$
 $B \times D = \{(1,5), (1,6), (2,5), (2,6), \dots\}$

Clearly every element of $A \times C$ is contained in $B \times D$. So, $A \times C \subset B \times D$

Question

Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Write $A \times B$. How many subsets will $A \times B$ have? List them.

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

Prove that

$$A \times B = B \times A$$

$$\Leftrightarrow A = B.$$

$$2^4 = 16.$$

$$+ \begin{matrix} & (a, b) \\ \hline \end{matrix}$$

$$\begin{aligned} & \phi, \{(1, 3)\}, \{(1, 4)\}, \{(2, 3)\} \\ & \{(2, 4)\}, \{(1, 3), (1, 4)\} \end{aligned}$$

Question

Let A and B be two sets such that $n(A) = 3$ and $n(B) = 2$. If $(x, 1), (y, 2), (z, 1)$ are in $A \times B$ find A and B , where x, y and z are distinct elements.

(Sol) Given $n(A) = 3, n(B) = 2$

$$\& (x, 1), (y, 2), (z, 1) \in A \times B$$

$$\Rightarrow x, y, z \in A \text{ and } 1, 2 \in B.$$

$$\text{So, } A = \{x, y, z\} \text{ and } B = \{1, 2\}.$$

Question

The Cartesian product $A \times A$ has 9 elements among which are found $(-1, 0)$ and $(0, 1)$. Find the set A and the remaining elements of $A \times A$.

$$n(A \times A) = 9$$

$$\Rightarrow n(A) \times n(A) = 9$$

$$\Rightarrow [n(A)]^2 = 9$$

$$\Rightarrow n(A) = \sqrt{9}$$

$$\Rightarrow n(A) = 3$$

$$(-1, 0), (0, 1) \in A \times A$$

$$\Rightarrow -1, 0, 1 \in A$$

$$\downarrow$$

$$A = \{-1, 0, 1\}$$

Thank
You

2.